

Connectivity flow — Squirro ↔ ToolUniverse SMCP

Ticket: [DSR-443](#) · **Scope:** sempart-demo + sr-dev, this commit ·

Audience: Squirro IT. Companion to `README.md` (ops).

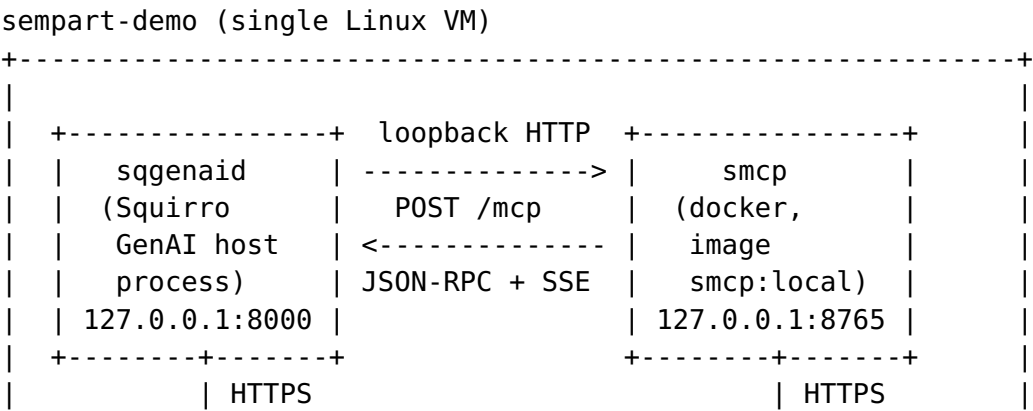
Two hosts, one image. SMCP is deployed on both `sempart-demo` and `sr-dev`. The container, egress surface (§3, Appendix A), and trust properties of the *server itself* are identical — same `smcp:local` image, same loopback bind. What differs is **how Squirro reaches it**: on `sempart` the GenAI client is a host process (shares host loopback); on `sr-dev` it is a container (`squirro-service-genai`) with its own loopback, so the client→server hop crosses a container boundary (§1.2).

1. Topology

On both hosts the SMCP container’s listener is bound to **host loopback only** (`127.0.0.1:8765`): only the host’s network namespace can reach it. No `nginx` between Squirro and SMCP. The difference between hosts is which side of a container boundary the Squirro MCP *client* sits on.

1.1 sempart-demo — GenAI is a host process

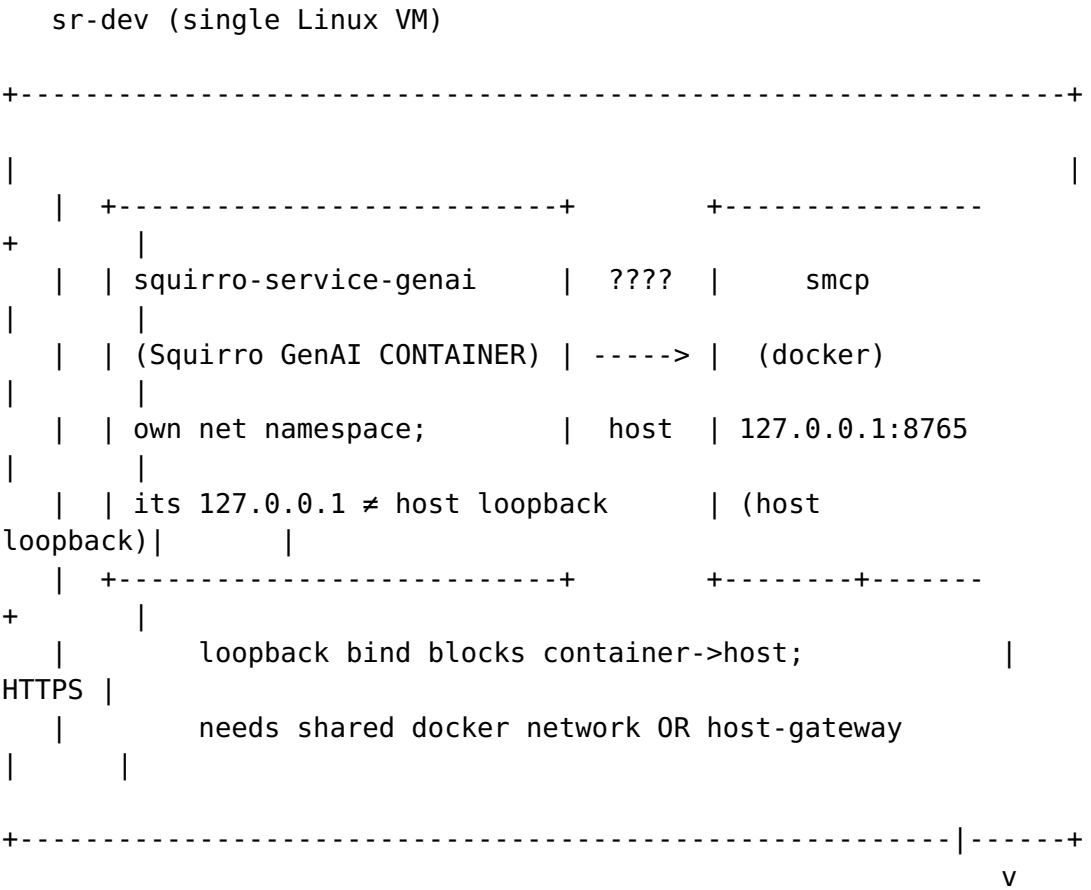
`sqgenaid` runs directly on the host, so it and the published SMCP port share the same loopback. The client→server hop is a plain loopback call.



+----- -----+ v		+----- -----+ v	
+-----+	also from	+-----+	
api.openai	<---- smcp -----	External biomed	
.com	(find_tools	APIs (§3)	
+-----+	ranking)	+-----+	
Address	Listener	Reachable from	Auth
127.0.0.1:8765/ mcp	smcp container (via 0.0.0.0:8000 inside)	Same host only	None (see §4)
127.0.0.1:8000	sqgenaid host process	Same host only	(Squirro internal)

1.2 sr-dev — GenAI is a container

Here Squirro GenAI runs as the `squirro-service-genai` **container**. A container has its own loopback, so `127.0.0.1:8765` *inside* that container is **not** the host's `127.0.0.1:8765` where SMCP is published. The plain loopback hop that works on sempart does **not** work here.



App. A)

Consequences for wiring Squirro → SMCP on sr-dev (one must hold before the client can reach the server; **none are configured yet** — the container is up and `host-side smoke.sh` passes, but the cross-boundary hop is the open integration item):

- 1. **Shared docker network** — attach `smcp` and `squirro-service-genai` to the same user-defined network; the client uses `http://smcp:8000/mcp` (service DNS). Keeps traffic off the host loopback entirely.
- 2. **Docker bridge gateway** — client targets the host gateway IP (e.g. `172.17.0.1:8765`) reachable from the container. Requires SMCP’s published port to accept the bridge source, widening the bind beyond strict `127.0.0.1` — re-examine §4 if taken.
- 3. **Host networking for smcp** — `network_mode: host`; then the container’s `127.0.0.1` is the host’s. Drops the port-mapping isolation; re-examine §4.

Option 1 is preferred: it keeps SMCP off any externally-routable address and preserves the §4 “no listener beyond what one host-local peer needs” posture, just scoped to a docker network instead of host loopback.

Address	Listener	Reachable from	Auth
127.0.0.1:8765/ mcp	smcp container (via 0.0.0.0:8000 inside)	Host namespace only — not the GenAI container as-is	None (see §4)
Squirro GenAI	squirro- service-genai container	(Squirro internal)	(Squirro internal)

2. On-the-wire sequence

Squirro’s MCP client (Studio-UI-configured at deploy time with URL `http://127.0.0.1:8765/mcp` on `sempart`; the `docker-network/gateway` form from §1.2 on `sr-dev` — no auth either way) speaks standard MCP streamable-HTTP. One init handshake per session, then `tools/call` per LLM tool invocation. The sequence below is identical on both hosts; only the client’s target URL differs.

```

sqgenaid                                smcp
OpenAI /                                (FastMCP, compact mode)
(MCP client)
biomed API
|
|
| |--- POST /mcp initialize ----->|
|
| |<-- 200 + mcp-session-id: <uuid>--|
|
|
| |--- POST /mcp notifications/      |
|
| |   initialized (+session hdr) -->|
|
| |<-- 202
|-----|
|
|
| |--- POST /mcp tools/list ----->|
|
| |<-- 200 + 5 meta-tools -----| // compact mode
returns: |
|
| | // find_tools,
list_tools,|
|
| | // grep_tools,
|
| | //
get_tool_info, |
|
| | //
execute_tool |
|
|
| | (5 schemas forwarded into agent's OpenAI tools[])
|
|
|
| |--- POST /mcp tools/call ----->|
|
| |   name=find_tools | ToolFinderLLM ranks tools
|
| |   args={query, limit} |--- HTTPS chat/completions-
>|
|
|
|-----|
| |<-- 200 + ranked tool descriptors-|
|
|
|
| |--- POST /mcp tools/call ----->|
|

```

```

|     name=execute_tool           | resolve, dispatch to TU
|
|     args={tool_name, arguments} | --- HTTPS to external API -
>|
|                               |
<-----|
|<-- 200 + tool output JSON -----|
|

```

Key facts:

- tools/list returns **5 meta-tools, not ~2,278**. Compact mode loads the full TU catalog in the background; the agent reaches breadth via tools/call name=execute_tool (smcp.py:1087, :1116).
- find_tools causes a **second outbound HTTPS to OpenAI from smcp** (for ranking), independent of the agent's own OpenAI call.
- The custom JSON-RPC methods tools/find / tools/search are declared but **not enabled** (middleware commented at smcp.py:558); discovery flows through the find_tools MCP tool above.

3. Egress destinations

Two containers on sempart make outbound HTTPS. The table below names the hosts a typical session hits; the **full inventory (~315 hosts) is in Appendix A** at the end of this document. .env.template lists API key variables but not hostnames.

Domain	Purpose	From	Auth
api.openai.com	LLM (agent + Tool_Finder_LLM ranking)	sqgenaid AND smcp	API key
api.platform.opentargets.org	Target ↔ disease ↔ drug graph	smcp	None
api.fda.gov	Drug labels, FAERS adverse events	smcp	Optional key
clinicaltrials.gov	Trial registry	smcp	None
rest.uniprot.org	Protein metadata	smcp	None
www.ebi.ac.uk	ChEMBL, EuropePMC,	smcp	None

Domain	Purpose	From	Auth
	Reactome, AlphaFold		
euutils.ncbi.nlm.nih.gov	PubMed, gene, taxonomy	smcp	Optional key
www.proteinatlas.org	HPA expression data	smcp	None
ops.epo.org	EPO patent search	smcp	OAuth2
Other biomedical APIs	DisGeNET, OMIM, OncoKB, BRENDA, USPTO, ICD, ...	smcp	Mixed; see .env.template

OpenAI is the only destination called by both containers — a firewall allow-list for *.openai.com is the single most critical egress rule. No proxy, no TLS pinning, no outbound rate limiting from our side.

4. Trust posture

- **No auth on the MCP endpoint.** Loopback bind (docker-compose.yml:14) is the sole access control on both hosts: reachability requires shell on the host (htafer, non-root, passwordless sudo, not in docker group). Changing to 0.0.0.0:8765 or fronting with unauthenticated nginx turns the server into an open relay (our OpenAI key + ~30 biomedical APIs). **sr-dev caveat:** wiring the GenAI *container* to SMCP via the bridge-gateway or host-networking options (§1.2 options 2/3) widens the bind past strict host loopback and weakens this control — prefer the shared-docker-network option (§1.2 option 1), which keeps SMCP off any host-routable address while still reaching the container.
- **Secrets** in smcp/.env: gitignored, on-host only, mounted via compose env_file: (not baked into the image). .env.template lists every variable; treat .env as the most sensitive file here.
- **Image** smcp:local is built on each host from the git checkout (squirrel/ToolUniverse fork, branch swiss-rockets, anonymous HTTPS pull). No registry push; the two hosts build the same source independently.
- **Logs:** json-file, 50 MB × 3 (docker-compose.yml:33-37). Tool arguments and results are **unredacted**. No central forwarding.

- **Blast radius if compromised:** OpenAI spend on our key, rate-limit consumption on biomedical APIs, exfiltration via tool output (read-only public data — no upstream write paths from the TU tools here).
- No privilege boundary between `sqgenaid` and `smcp` beyond host process isolation; they share trust class.

5. Deployment status

This document covers **sempart-demo** + **sr-dev**. Status across the clusters in scope for the broader DSR-394 work:

- **sempart-demo** (demo): SMCP **deployed**, container healthy, host-side `smoke.sh` green. Squirro→SMCP wired via host loopback (§1.1).
- **sr-dev** (acceptance): SMCP container **deployed**, healthy, host-side `smoke.sh` green. **Squirro→SMCP not yet wired** — the GenAI container cannot reach host loopback; needs the §1.2 cross-boundary route (prefer shared docker network). Studio-UI registration is additionally blocked by nginx (POST `/studio/*`), so register via API/config, not the UI.
- **sr-dev-com**: SMCP not deployed.
- **swiss-rockets.squirro.com** (production): SMCP not deployed. Any production rollout that exposes SMCP beyond loopback requires auth, TLS, and rate limiting before the §4 trust model holds — treat as a new connectivity review.

6. References

- README.md — operational deploy/smoke instructions.
- `docker-compose.yml`, `Dockerfile`, `.env.template` — runtime config.
- `smoke.sh` — minimal client showing the handshake on the wire.
- `libs/tooluniverse/src/tooluniverse/smcp.py:1087` — `_expose_core_discovery_tools` (the 4 dispatch meta-tools).
- `libs/tooluniverse/src/tooluniverse/smcp.py:1116` — `_add_search_tools` (registers `find_tools`).
- `libs/tooluniverse/src/tooluniverse/smcp.py:558` — `tools/find/tools/search` middleware (currently disabled).
- MCP protocol: <https://modelcontextprotocol.io/>

7. Appendix A — Full egress endpoint inventory

Generated by grepping every `url / endpoint / base_url` field in `src/tooluniverse/data/**/*.json` and every URL constant or requests.`{get,post,...}` call in `src/tooluniverse/**/*.py` of the deployed commit. Plus a small set of **SDK-default endpoints** that don't appear as string literals but are reached via the LLM client libraries (openai, google-generativeai, huggingface_hub).

~**315 distinct hosts**. Any tool's invocation through `execute_tool` may reach the corresponding host; compact-mode loads the full TU catalog at startup, so every host below is *reachable* even though the SR persona's queries typically hit only the §3 subset.

A.0 Wildcard summary (for firewall allow-listing)

Apex / wildcard	Host count	Purpose
*.openai.com	(SDK default)	LLM — agent + Tool_Finder_LLM ranking
*.nlm.nih.gov	14	NCBI / NIH services (PubMed, PubChem, RxNorm, UMLS, MedlinePlus, ...)
*.ebi.ac.uk	2	EMBL-EBI hub (ChEMBL, Europe PMC, ENA, GxA, GWAS, Intact, InterPro, PRIDE, PDBe, Rhea, ...)
*.nih.gov	8	Other NIH (NIA, NCI, NCATS, GDC, NCBO, ...)
*.who.int	4	ICD-11, GHO
*.opentargets.org	2	Open Targets Platform + Genetics
*.rcsb.org	3	PDB structures, files, search
*.open-meteo.com	5	

Apex / wildcard	Host count	Purpose
		Weather/climate APIs (non-biomedical)
*.nvidia.com	4	NVIDIA NIM / BioNeMo inference
*.hubmapconsortium.org	3	HuBMAP single-cell
*.clinicalgenome.org	4	ClinGen / ERepo / Actionability
*.nasa.gov	4	NASA reference (non-biomedical)
*.huggingface.co	(SDK default)	HF model hub + inference

If the firewall does **not** support wildcard rules, allow-list every host listed in §A.1–A.9 explicitly.

A.1 LLM and inference providers

Called from smcp for Tool_Finder_LLM (and from sqgenaid itself for the agent’s own reasoning). **High-traffic and billable.**

```

api.openai.com           # OpenAI (SDK default; primary LLM)
*.openai.azure.com       # Azure OpenAI (host depends on AZURE_OPENAI_ENDPOINT)
generativelanguage.googleapis.com # Google Gemini (SDK default)
openrouter.ai           # OpenRouter fallback provider
huggingface.co          # HF Hub: models + embeddings
api-inference.huggingface.co # HF Inference API (SDK default)
api.nvcf.nvidia.com      # NVIDIA NIM cloud functions
build.nvidia.com         # NVIDIA build endpoints
integrate.api.nvidia.com  # NVIDIA inference
health.api.nvidia.com    # NVIDIA biomedical NIMs (BioNeMo)
assets.ngc.nvidia.com    # NVIDIA NGC model assets
azure-ai.hms.edu         # Harvard Medical School Azure
helper (smolagent_tools)

```

A.2 NCBI / NLM (US National Library of Medicine)

```

api.ncbi.nlm.nih.gov     # NCBI Datasets v2
eutils.ncbi.nlm.nih.gov  # E-utilities (PubMed, gene, taxonomy, sequence)
pubmed.ncbi.nlm.nih.gov  # PubMed UI fallback
pmc.ncbi.nlm.nih.gov     # PubMed Central

```

www.ncbi.nlm.nih.gov	# NCBI web (LitVar, PubTator3, IntAct mirror)
pubchem.ncbi.nlm.nih.gov	# PubChem (compounds, substances, bioassays)
trace.ncbi.nlm.nih.gov	# NCBI SRA Trace
clinicaltables.nlm.nih.gov	# NLM clinical lookup tables
dailymed.nlm.nih.gov	# DailyMed drug labels
id.nlm.nih.gov	# NLM identifier service
rxnav.nlm.nih.gov	# RxNorm / drug nomenclature
uts.nlm.nih.gov	# UMLS Terminology Services
uts-ws.nlm.nih.gov	# UMLS web service
wsearch.nlm.nih.gov	# NLM web search

A.3 Other NIH and US-gov biomedical

api-evsrest.nci.nih.gov	# NCI EVS terminology
api.gdc.cancer.gov	# NCI Genomic Data Commons
api.nal.usda.gov	# USDA FoodData Central
cactus.nci.nih.gov	# NCI Chemical Identifier Resolver
data.cdc.gov	# CDC open data
fdc.nal.usda.gov	# USDA FoodData Central (alt)
healthabc.nia.nih.gov	# NIA Health ABC study
icite.od.nih.gov	# NIH iCite (citation analysis)
ontology.api.hubmapconsortium.org	
entity.api.hubmapconsortium.org	
search.api.hubmapconsortium.org	
odphp.health.gov	# HHS ODPHP guidelines
pdc.cancer.gov	# NCI Proteomic Data Commons
pharos-api.ncats.io	# NCATS Pharos
pharos.nih.gov	# NCATS Pharos web
run.opencravat.org	# NCATS OpenCRAVAT
services.cancerimagingarchive.net	# TCIA
webapis.cancer.gov	# NCI Drug Dictionary
www.atsdr.cdc.gov	# CDC ATSDR
www.cdc.gov / wwwn.cdc.gov	# CDC reference
www.nia.nih.gov	# NIA

A.4 EMBL-EBI and Europe PMC

www.ebi.ac.uk is a single host serving dozens of TU tools: ArrayExpress, BioModels, BioStudies, ChEMBL, dbfetch, EBI Search, EFO, EMDB, ENA browser, Europe PMC Citations, Europe PMC, eQTL, EVA, Gene2Phenotype, GWAS Catalog, IntAct, InterPro, MetaboLights, MGnify, OLS, PDBe (API + Graph), PRIDE, Proteins API.

www.ebi.ac.uk	
alphafold.ebi.ac.uk	# AlphaFold protein structures
europepmc.org	# Europe PMC literature
rfam.org / batch.rfam.org	# RNA family DB (EBI-hosted)

tess.elixir-europe.org	# ELIXIR Tess training portal
jaspar.elixir.no	# JASPAR TF binding profiles

A.5 Major curated biomedical platforms

api.platform.opentargets.org	# Open Targets (target-disease-drug)
api.genetics.opentargets.org	# Open Targets Genetics
rest.uniprot.org	# UniProt
rest.ensembl.org	# Ensembl REST
rest.kegg.jp	# KEGG pathways/compounds
data.rcsb.org	# PDB metadata
files.rcsb.org	# PDB structure files
search.rcsb.org	# PDB search
reactome.org	# Reactome pathways
plantreactome.gramene.org	# Plant Reactome
string-db.org	# STRING protein interactions
stitch.embl.de	# STITCH compound-protein
dgidb.org / www.dgidb.org	# DGIdb drug-gene interactions
www.proteinatlas.org	# HPA Human Protein Atlas
api.cellxgene.cziscience.com	# CZI cellxgene Census
api.brain-map.org	# Allen Brain Atlas
api.geneontology.org	# Gene Ontology
golr-aux.geneontology.io	# GO golr
api.monarchinitiative.org	# Monarch Initiative
pantherdb.org	# PANTHER classification
public.ndexbio.org	# NDEx networks
clue.io / api.clue.io	# Broad CMap / Connectivity Map

A.6 Genomic variants and clinical genetics

gnomad.broadinstitute.org	# gnomAD population variation
rest.variantvalidator.org	# VariantValidator HGVS
mutalyzer.nl	# Mutalyzer HGVS
panelapp.genomicsengland.co.uk	# Genomics England PanelApp
api.disgenet.com	# DisGeNET (auth required)
www.disgenet.org	# DisGeNET web
beacon.progenetix.org	# GA4GH Beacon (Progenetix)
search.clinicalgenome.org	
reg.clinicalgenome.org	
erepo.clinicalgenome.org	
actionability.clinicalgenome.org	
search.thegencc.org	
thegencc.org	
wintervar.wglab.org	# InterVar
cancervar.wglab.org	# CancerVar
regulomedb.org	# RegulomeDB
databases.lovd.nl	# LOVD locus-specific
api.mavedb.org	# MaveDB
r12.finnngen.fi	# FinnGen

www.internationalgenome.org # 1000 Genomes
www.deciphergenomics.org # DECIPHER
www.phenoscanter.medschl.cam.ac.uk # PhenoScanner
cadd.bihealth.org # CADD scoring
cadd.zju.edu.cn # CADD (CN mirror)
pangolin-37-xwkwwwxdwq-uc.a.run.app # Pangolin splice predictor (GCR)
pangolin-38-xwkwwwxdwq-uc.a.run.app
spliceai-37-xwkwwwxdwq-uc.a.run.app # SpliceAI (GCR)
spliceai-38-xwkwwwxdwq-uc.a.run.app
alphamissense.hegelab.org # AlphaMissense (hegelab mirror)
hocomocol4.autosome.org # HOCOMOCO TF motifs
api.genome.ucsc.edu # UCSC Genome
data.4dnucleome.org # 4D Nucleome
data.orthodb.org # OrthoDB
omabrowser.org # OMA orthologs
gtdb-api.ecogenomic.org # GTDB taxonomy
gtexportal.org # GTEx eQTL
www.encodeproject.org # ENCODE
chip-atlas.org # ChIP-Atlas
www.intogen.org # IntOGen cancer drivers
www.cbioportal.org # cBioPortal
www.dfam.org # DFAM repeats
www.genomenexus.org # Genome Nexus

A.7 Chemistry, drugs, pharmacology, screening

www.bindingdb.org # BindingDB
www.guidetopharmacology.org # IUPHAR/BPS Guide to Pharmacology
www.swissadme.ch # SwissADME
swissdock.ch / www.swissdock.ch # SwissDock
www.swisstargetprediction.ch # SwissTargetPrediction
www.lipidmaps.org / lipidmaps.org # LIPID MAPS
www.metabolomicsworkbench.org # Metabolomics Workbench
metabolomics-usi.gnps2.org # GNPS metabolomics
massbank.eu # MassBank mass spectra
massive.ucsd.edu # UCSD MassIVE
proteomecentral.proteomexchange.org
proteins.plus # ProteinsPlus
klifs.net # KLIFS kinase structures
zinc15.docking.org # ZINC15
mcule.com / doc.mcule.com # Mcule virtual chemistry
enamine.net / new.enaminestore.com
api.emolecules.com / www.emolecules.com
www.t3db.ca # T3DB toxins
www.drugbank.ca # DrugBank XML
hmdb.ca / www.hmdb.ca # HMDB Human Metabolome
www.rhea-db.org # Rhea reactions
enzyme.expasy.org # ExPASy enzyme
prosite.expasy.org # PROSITE
www.idtdna.com # IDT oligo design

www.crystallography.net	# COD crystal structures
tmapi.neb.com	# NEB enzyme finder
mibig.secondarymetabolites.org	# MIBiG secondary metabolites
www.tcdb.org	# TCDB transporter classification
www.gsea-msigdb.org	# MSigDB
elm.eu.org	# ELM eukaryotic linear motifs
www.openml.org	# OpenML ML datasets
biit.cs.ut.ee	# g:Profiler
db.idrblab.net	# IDRB drug-target
drugcentral.org	# DrugCentral
hb.flatironinstitute.org	# Flatiron H-bond DB
db.indra.bio	# INDRA pathway DB

A.8 Pharmacogenomics and clinical guidelines

api.cpicpgx.org / cpicpgx.org	# CPIC pharmacogenomic guidelines
api.clinpgx.org	# ClinPGx
api.pharmgkb.org / www.pharmgkb.org	# PharmGKB
www.pharmvar.org	# PharmVar
www.nice.org.uk	# NICE guidelines (UK)
www.nccn.org	# NCCN guidelines (US)
www.sign.ac.uk	# SIGN guidelines (Scotland)
canadiantaskforce.ca	# Canadian Task Force
guidelines.ebmportal.com	
api.magicapp.org	# MAGICapp digital guidelines
www.tripdatabase.com	# TRIP evidence database

A.9 Clinical trials, regulatory, drug safety

clinicaltrials.gov	# NIH ClinicalTrials.gov
api.fda.gov	# openFDA (FAERS, drug labels)
open.fda.gov / www.fda.gov	# openFDA web
sideeffects.embl.de	# SIDER adverse effects
www.oncokb.org / api.oncokb.org	# OncoKB
icd.who.int	# WHO ICD-11
icdaccessmanagement.who.int	
id.who.int / www.who.int	# WHO
ghoapi.azureedge.net	# WHO Global Health Observatory
disease.sh	# disease.sh COVID-19
nextstrain.org	# Nextstrain phylogenetics
www.disease-ontology.org	# Disease Ontology
ontology.jax.org	# JAX HPO
oncotree.mskcc.org	# MSKCC OncoTree
api.orphancode.org	# Orphanet codes
api.orphadata.com	# Orphadata rare disease
api.omim.org	# OMIM (auth required)

A.10 Patents and IP

ops.epo.org	# EPO Open Patent Services
(OAuth2)	
api.uspto.gov	# USPTO API

A.11 Protein structure, function, interactions

alphafold.ebi.ac.uk	# AlphaFold (also in §A.4)
mobidb.org / mobidb.bio.unipd.it	# MobiDB disordered regions
www.disprot.org	# DisProt
swissmodel.expasy.org	# SWISS-MODEL
www.cathdb.info	# CATH structural classification
www.sasbdb.org	# SASBDB SAXS
channelsdb.ncbr.muni.cz	# ChannelsDB
pdb-redo.eu	# PDB-REDO refinement
scop.mrc-lmb.cam.ac.uk	# SCOP
www.imgt.org	# IMGT immunoglobulins
www.fpbase.org	# FPbase fluorescent proteins
www.proteomicsdb.org	# ProteomicsDB
www.pathwaycommons.org	# Pathway Commons
opig.stats.ox.ac.uk	# OPIG (SAbDab, etc.)
apprisws.bioinfo.cnio.es	# APPRIS principal isoforms
deepgo.cbrc.kaust.edu.sa	# DeepGO function prediction
research.bioinformatics.udel.edu	# ProteoCons
evemodel.org	# EVEmodel
zitniklab.hms.harvard.edu	# Zitnik lab tools (Harvard)

A.12 Expression, single-cell, cell atlases, omics

api.cellxgene.cziscience.com	# cellxgene Census (also §A.4)
service.azul.data.humancellatlas.org	# HCA
maayanlab.cloud	# Maayan lab tools (ARCHS4, ChEA, ...)
www.bgee.org	# Bgee gene expression
www.pombase.org	# PomBase yeast
www.yeastgenome.org	# SGD yeast
www.flymine.org / www.mousemine.org / www.humanmine.org	# InterMine
targetmine.mizuguchilab.org	# TargetMine
bigg.ucsd.edu	# BiGG genome-scale models
api.cellmodelpassports.sanger.ac.uk	
api.cellosaurus.org	# Cellosaurus cell lines
www.alliancegenome.org	# Alliance of Genome Resources
rest.genenames.org	# HGNC gene names
www.mousephenotype.org	# IMPC
rest.wormbase.org	# WormBase
rest.rgd.mcw.edu	# Rat Genome DB (in source, schema-only)
plantreactome.gramene.org	# Plant Reactome

synergxdb.ca	# SynergxDB drug combos
veupathdb.org	# VEuPathDB (EuPathDB)
neuromorpho.org	# NeuroMorpho
neurovault.org	# NeuroVault
modeldb.science	# ModelDB neural models
idr.openmicroscopy.org	# IDR imaging
sabiork.h-its.org	# SABIO-RK kinetics
query-api.iedb.org	# IEDB immune epitopes
www.immport.org (schema)	# ImmPort
vdjdb.com	# VDJdb TCR
omnipathdb.org	# OmniPath signaling
gpcrdb.org	# GPCRdb
mychem.info	# MyChem.info
mydisease.info	# MyDisease.info
mygene.info	# MyGene.info
myvariant.info	# MyVariant.info
bio-bigdata.hrbmu.edu.cn	# HRBMU multi-omics
biocyc.org / webservice.biocyc.org	# BioCyc / MetaCyc
ctdbase.org	# CTD chemical-gene-disease
www.disprot.org	# DisProt (also §A.11)
rnacentral.org	# RNACentral

A.13 Ontologies, registries, ID resolvers

purl.obolibrary.org	# OBO Foundry ontologies
bioregistry.io	# Bioregistry
registry.api.identifiers.org	
resolver.api.identifiers.org	
data.bioontology.org	# NCB0 BioPortal data
bioportal.bioontology.org	# NCB0 BioPortal
bio.tools	# ELIXIR bio.tools registry
biotools_registry (alias)	
api.catalogueoflife.org	# Catalogue of Life
api.ror.org / ror.org	# Research Organization Registry
api.crossref.org	# Crossref DOI
api.datacite.org	# DataCite DOI
doi.org	# DOI resolver
pub.orcid.org	# ORCID public API
www.re3data.org	# re3data repository registry
identifiers.org (legacy)	
www.itis.gov	# ITIS taxonomy

A.14 Literature, citations, scholarly identifiers

arxiv.org / export.arxiv.org	# arXiv
api.biorxiv.org	# bioRxiv
api.archives-ouvertes.fr	# HAL
api.core.ac.uk / core.ac.uk	# CORE
api.openaire.eu	# OpenAIRE
api.openalex.org / openalex.org	# OpenAlex

api.opencitations.net	# OpenCitations
api.scite.ai	# Scite
api.semanticscholar.org / www.semanticscholar.org	# Semantic Scholar
api.unpaywall.org	# Unpaywall
dblp.org	# DBLP CS bibliography
doaj.org	# DOAJ open-access journals
inspirehep.net	# INSPIRE-HEP
scholar.archive.org	# Internet Archive Scholar
datadryad.org	# Dryad data
dataverse.harvard.edu	# Harvard Dataverse
api.figshare.com	# Figshare
api.osf.io	# Open Science Framework
zenodo.org	# Zenodo
api.developers.addgene.org	# Addgene plasmids
ec.europa.eu	# EC open data

A.15 Ecology, biodiversity, taxonomy

api.gbif.org	# GBIF biodiversity occurrences
api.inaturalist.org	# iNaturalist
api.obis.org	# OBIS ocean biodiversity
api.ebird.org	# eBird
api.opentreeoflife.org	# Open Tree of Life
api.census.gov reference)	# US Census (also non-bio)
eol.org	# Encyclopedia of Life
paleobiodb.org	# Paleobiology DB
search.idigbio.org	# iDigBio specimens
marineregions.org / www.marineregions.org	
www.marinespecies.org	# WoRMS marine species
powo.science.kew.org	# Kew Plants of the World
nextstrain.org	# phylogenetics (also §A.9)
synbiohub.org	# SynBioHub
www.openml.org	# OpenML (also §A.7)

A.16 Earth, weather, space, non-biomedical reference

Loaded into TU but **never invoked by the SR biomedical persona** in normal operation. Listed for completeness.

api.weather.gov	# NWS US weather
api.open-meteo.com	# Open-Meteo
air-quality-api.open-meteo.com	
archive-api.open-meteo.com	
climate-api.open-meteo.com	
flood-api.open-meteo.com	
geocoding-api.open-meteo.com	
marine-api.open-meteo.com	
api.met.no	# MET Norway

api.waqi.info	# World Air Quality Index
api.opentopodata.org	# OpenTopoData elevation
api.sunrise-sunset.org	
nominatim.openstreetmap.org	# OSM geocoding
earthquake.usgs.gov	# USGS earthquakes
waterservices.usgs.gov	# USGS water
cmr.earthdata.nasa.gov	# NASA Earthdata
eonet.gsfc.nasa.gov	# NASA EONET events
kauai.ccmc.gsfc.nasa.gov	# NASA CCMC space weather
ssd.jpl.nasa.gov / ssd-api.jpl.nasa.gov	# JPL Solar System Dynamics
exoplanetarchive.ipac.caltech.edu	
ned.ipac.caltech.edu	# NASA NED astronomy
simbad.cds.unistra.fr	# SIMBAD astronomy
skyserver.sdss.org	# SDSS skyserver
coastwatch.pfeg.noaa.gov	# NOAA ERDDAP
rest.isric.org	# ISRIC SoilGrids
www.worldbank.org / api.worldbank.org	# World Bank
catalogue.ceda.ac.uk	# CEDA climate archive
world.openfoodfacts.org	# Open Food Facts
www.countyhealthrankings.org	# County Health Rankings

A.17 Code, package registries, generic web

api.github.com	# GitHub (some agentic tools)
api.anaconda.org	# Anaconda.org packages
bioc.r-universe.dev	# Bioconductor R-universe
cran.r-pkg.org	# CRAN database
pypi.org / pypistats.org	# PyPI
en.wikipedia.org / www.wikipedia.org	
www.wikidata.org / query.wikidata.org / dbpedia.org	
ec.europa.eu	# EC OpenData
httpbin.org	# diagnostic (some test tools)

A.18 Out-of-cluster TU helpers (worth flagging)

These are TU calling **third-party-hosted TU services**, not under SR control:

```
tooluniversemcpsrver.onrender.com # Render-hosted TU instance –
`agentic_tools.json` references this as an external TU
                                # if invoked, smcp makes an
outbound HTTPS call to this third party
azure-ai.hms.edu                # Harvard Medical School Azure
helper – used by `smolagent_tools`
```

agentic_tools.json declares ~30 “RemoteTool” definitions pointing at tooluniversemcpsrver.onrender.com. They are loaded into the registry in compact mode. If the agent picks one of these via find_tools, our

server makes an outbound HTTPS call to Render-hosted infrastructure. For a hardened deployment, consider excluding `agentic_tools` (add to `--exclude-tools` in Dockerfile).

A.19 Hosts `grep'd` but not treated as endpoints

The static analysis also surfaced ~200 documentation URLs (e.g. `*.readthedocs.io`, `pytorch.org`, `numpy.org`, `scikit-learn.org`, `matplotlib.org`, `flask.palletsprojects.com`, etc.) embedded in tool description text. These are **not invoked**; they appear because tool metadata cites the upstream library's docs. Not included above.

Methodology notes

- Inventory generated against `libs/tooluniverse/` at the deployed commit.
- The deployed container loads the full TU catalog via `--compact-mode` but excludes `Tool_RAG` (see Dockerfile:54). Excluding `Tool_RAG` does not change the egress surface — that tool is local-model based, not network-bound, so its absence has no effect on this list.
- The hostname list is what the deployed image **can** reach if the matching tool is invoked. The §3 table is what it **typically** reaches during normal `sempart-demo` use.
- For an ongoing watch, re-run: `grep -rhoE 'https?://[a-zA-Z0-9._-]+\.[a-zA-Z]{2,}' libs/tooluniverse/src/tooluniverse/ | sort -u` and diff against the list above after TU upstream merges.